

Bachelor in Computer Applications

Course Title: **C Programming**

Code No: Semester: **1**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Explain the role of function in programming. How function is declared, defined and called in C? Write a program to find factorial of 'n' using recursive function.
2. What is an Identifier and keyword? Explain the rules for naming valid identifiers in C with example. Write a C program to find sum of digits of a 'n' digit number.
3. How structure is different from union? Write a program to store and display basic information (roll, name, address, email, and phone) of students of using a structure.
4. What is the basic structure of a C program? Explain each part.
5. How break statement is different from continue statement. Explain with examples?
6. What do you mean by multi-dimensional array? Write program logic to add and display the sum of two m x n matrices.
7. Find errors (If any) in the following program and rewrite it.
`#define (studio.h)float main < >int m [ ] [ ] = {{1,2,3,4}};for (i = 4, l>=0 ; i--)for (j=0, j <4; j++);m [i] [j] = a[i][j] + b[i][j];return 0. 0;}`
8. Write a program to generate the following pattern of Integers.
1 1 2 1 2 3 1 2 3 4 1 2 3 4 5
9. Write a C program to check whether a given string is palindrome or not.
10. What is pointer? Illustrate the use of a double pointer with suitable example.
11. Why file is used in programming? Explain different file opening modes.
12. Write short notes on:
a. Macro
b. Null Pointer